



Monetary Authority
of Singapore

Orchid Blueprint

Infrastructure for safe and innovative use of digital
money in Singapore

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Version	Date	Author	Rationale
1.0	Nov 2023	Monetary Authority of Singapore	First publication

1 Introduction

Technological advancements have given rise to the possibility for commerce to be conducted without requiring transacting parties to meet in person. Electronic payments methods in Singapore are pervasive and continue to grow in terms of ease and convenience. Banking services are increasingly offered digitally, with efforts to increase their resiliency and reliability underway. Meanwhile, globally technological innovations in money are gaining traction, with central banks and commercial banks spending significant resources researching future forms of money and its impact on the economy.

With increased digitalisation, central bank digital currencies (CBDCs), tokenised bank liabilities (TBLs) and well-regulated stablecoins, together with a set of well-designed smart contracts, could grow in importance as a medium of exchange. These tokenised forms of digital money could support enhanced features that are not widely available today, including atomic settlement and the ability for senders to specify conditions, such as validity periods, when making transfers.

In view of these developments, MAS launched Project Orchid¹ in 2021 in collaboration with industry partners to explore the technical feasibility and infrastructure required for a digital Singapore dollar. The project started by examining potential use cases for digital money and the infrastructure that would be required to support it. In Phase 1, MAS collaborated with financial institutions, corporates, and government agencies and explored use cases to understand the potential opportunities and drawbacks of introducing a digital Singapore dollar. Phase 1 also introduced Purpose Bound Money (PBM) as a common protocol to specify conditions for digital money.

Phase 2 of the project undertook a two-pronged approach – first, the conduct of multiple industry-led digital money pilots that expanded the use of PBM across new scenarios; second, the formation of an industry group to examine how digital money might be effectively incorporated into Singapore’s payments and financial system.

This paper presents a blueprint (referred to as ***Orchid Blueprint***) which details the infrastructure for the safe and innovative use of digital money in Singapore, drawing on the findings from ongoing experiments and feedback from the industry group. The paper first outlines the motivations behind the introduction of the Orchid Blueprint, then provides an overview of the different forms of digital money. Next the paper delves into the technology infrastructure and standards required for digital money issuance and usage in Singapore. The paper then highlights the operational and commercial considerations for digital money and concludes by highlighting the areas that needs to be further explored.

¹ Project Orchid is a collaborative project between MAS and industry partners that aims to build the foundational digital infrastructure and blueprint required for a future digital money ready platform. collaboration.

2 Background and motivation

Advancements in digital money are largely driven by the rapid rise of enabling technologies such as tokenisation and distributed ledger technology (DLT), as well as growing interests in its purported benefits. Yet, it is prudent to balance the excitement of innovation with the safeguards of governance, regulation, and commercial viability.

The Orchid Blueprint bridges this gap by focusing on the key infrastructure components for safe and innovative use of digital money in Singapore. It also draws on the insights obtained from industry trials conducted under Project Orchid and inputs from the industry workstreams. This paper was developed with the perspective that it is valuable for central banks and the private sector to collaborate on matters concerning digital money.

The publication of an industry-wide blueprint helps to ensure interoperability across different infrastructures, platforms, and services. Without a coordinated approach at the outset, there would be unnecessary duplication of efforts, resulting in each provider needing to invest significantly to ensure compatibility and integration across different systems subsequently. Furthermore, the development of multiple proprietary infrastructure could result in inefficiencies and lead to higher costs for consumers and businesses, who might have to utilise several systems and incur additional fees from intermediaries to access financial services from different providers.

The emergence of different forms of digital money introduces its own set of risks, which require careful calibration of existing operating models by the central bank and policymakers. In addition, new forms of digital money should be interoperable with existing forms of money, to avoid fragmentation of payment systems in Singapore.

The Orchid Blueprint incorporates design considerations to facilitate clarity across the people, processes and technology required for the sustainable and responsible innovation in digital money. This paper focuses on the technological infrastructure and standards, with some coverage of a scheme that would define governance frameworks and commercial arrangements.

3 Forms of Digital Money

The future financial landscape is likely to feature the co-existence of multiple tokenised forms of digital money. These could include regulated stablecoins, tokenised bank liabilities (TBLs) and central bank digital currencies (CBDCs). The Orchid Blueprint is concerned with these forms of digital money, as described by the following:

Regulated Stablecoins

The Orchid Blueprint will support single-currency stablecoins that will be regulated for their value stability under MAS' proposed regulatory framework² for stablecoin activities. Stablecoins regulated under the framework will be recognised as an "MAS-regulated stablecoin". Any other stablecoin that is not regulated under the framework will not be supported under the Orchid Blueprint.

Tokenised Bank Liabilities

The Orchid Blueprint will support TBL that are issued by banks regulated by the MAS. TBLs refer to tokens issued by licenced banks that represent a general claim against the bank (i.e., backed by the issuing bank's balance sheet). Accordingly, TBLs differ from stablecoins that are backed by a pool of reserve assets (asset-backed). Licenced banks might issue TBLs with the aim of developing innovative new payment instruments or improving banking services. Existing licensing and prudential requirements, such as those under the Banking Act, will continue to apply in respect of banks issuing TBLs in Singapore. MAS will continue to monitor international regulatory developments and banks' initiatives in this space and impose additional requirements on TBLs as necessary.

Central Bank Digital Currency (CBDC)

The Orchid Blueprint will support CBDC that are issued by the MAS. CBDCs fall into two categories: (1) Retail CBDCs which are accessible by members of the public, just like cash is today; (2) Wholesale CBDCs whose access and use would be restricted to a select group of financial institutions, for the settlement of large-value interbank transactions.

² MAS' stablecoin regulatory framework seeks to ensure a high degree of value stability for stablecoins regulated in Singapore. Only single-currency stablecoins that are pegged to the Singapore Dollar or any G10 currency, and which are issued in Singapore, can apply to be regulated under the framework and be recognised as "MAS-regulated stablecoins", when the legislative amendments take effect. (MAS, 2023)

4 Key infrastructure components

The Orchid Blueprint defines the building blocks for digital money infrastructure in Singapore and describes how the use of digital money in banking and payments could be taken forward. The use of digital money for cross border trade and capital markets, such as trade execution and securities settlement, is currently outside the scope of this paper.

The building blocks describe the technical infrastructure and services involved in building the digital money capabilities. These building blocks play a key role in ensuring that digital money transfers in Singapore remain safe and efficient, even as new technologies are introduced.

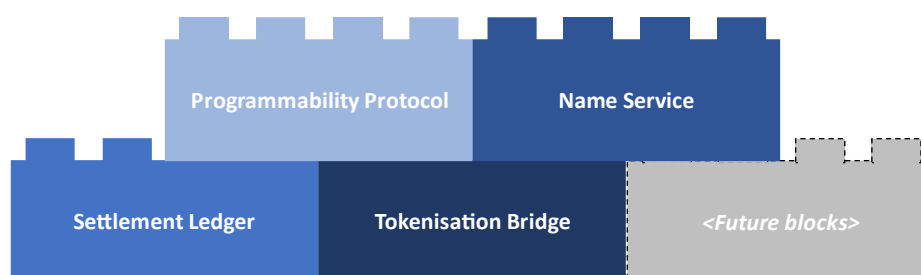


Figure 1: Building Blocks of the Orchid Blueprint

A brief description of the identified building blocks are as follows:

1. **Settlement Ledger** – A system for recording digital money transfers, with supporting features such as native programmability and atomic settlement of digital tokens.
2. **Tokenisation Bridge** – Connect existing account-based settlement systems with ledgers compatible with tokenised forms of digital money.
3. **Programmability Protocol** – Using Purpose Bound Money (PBM) as a common protocol to specify the conditions for the use of digital money.
4. **Name Service** – A service to translate between unwieldy wallet addresses and alternative name identifiers that are readable and meaningful for verification.
5. **Future Blocks** – Building blocks are designed in a modular fashion, allowing new technologies to be added as required to enhance digital money capabilities.

4.1 Settlement Ledger

Existing market infrastructures are responsible for facilitating the settlement of payments, which can be done on a gross or net basis. As they are responsible for facilitating the transfer of value and must ensure settlement finality, these settlement systems serve a systemically important role in the financial and economic landscape of Singapore. Accordingly, they are held to a high standard of performance and resilience³.

Similarly, settlement systems supporting digital money transfers are expected to uphold a high standard of performance and resilience. These may be seen similarly as financial market infrastructures (FMI), which consequently may fall under the scope of the application of the Principles

³ The Singapore: Financial Sector Assessment Program by IMF found that MEPS+ participants view the system as “reliable, resilient and fit-for-purpose” (IMF, 2019)

for Financial Market Infrastructures (PFMI)⁴, to the extent that these systems become systemically important and perform a function that an FMI does.

Under the existing account-based model, settlement occurs between entities who have accounts on a common ledger. A fund transfer is executed by debiting the originating account and crediting the beneficiary account by the corresponding amount. Information is exchanged between transacting parties, with the requirement that transacting parties must have accounts with attestable and compliant information to facilitate the funds transfer.

Meanwhile, in the context of token-based systems, a digital token represents a claim against its issuer. These tokens could be distributed as a digital bearer asset and transferred between the transacting parties. The settlement system here consists of the digital tokens and the settlement ledger which records the transfers of the digital token. Hence, a distinction is made between the ledger - which records ownership of the asset and the medium of exchange- the digital token itself. Settlement ledger in the context of the Orchid Blueprint refers primarily to the infrastructure by which the ownership and transfer of these tokens is recorded.

Although settlement ledgers are typically implemented based on DLT, the concepts discussed in the context of this Orchid Blueprint are designed to be applicable to non-DLT ledgers too.

Augmenting existing market infrastructure

Settlement ledgers can augment existing market infrastructure such as MEPS+ and bring about new capabilities including increased availability (such as 24/7 availability), native programmability, and atomic settlement. Due to the nature of digital money transfers that are conducted on settlement ledgers, these ledgers simultaneously serve the functions of payment, clearing and settlement systems⁵.

The supporting settlement ledger should enable multiple financial institutions to host financial applications, issue digital assets (including digital money and other tokenised assets) and support both commercial and government use cases.

Multilateral platforms

Current developments suggest that it is unlikely there will be one common platform upon which members around the world would agree to transact on. Instead, it is envisioned that the global financial landscape will be proliferated with many independent platforms, each with their own governance structure and serving their own constituents.

Domestically, the circumstances are different. Given the existence of a common legal and regulatory framework, the probability of converging towards a unified settlement ledger to achieve economies of scale is higher. For example, in Singapore, FAST (Fast and Secure Transfers)⁶ currently serves as a common fund transfer service between participating commercial banks and non-bank financial institutions (NFI) that allows customers of one entity to transfer Singapore dollar funds to another almost instantly. The establishment of a domestic settlement ledger would require participating

⁴ The PFMI define an FMI as a multilateral system among participating institutions, including the operator of the system, used for the purposes of clearing, settling or recording payments, securities, derivatives or other financial transactions. (BIS, 2023).

⁵ As defined in the published report on Payment, Clearing and Settlement Systems in Singapore (MAS, 2011)

⁶ More information about FAST may be obtained from the Association of Banks in Singapore (ABS) at <https://www.abs.org.sg/consumer-banking/fast> (ABS, 2014)

financial institutions to be aligned on the commercial and operating model. This is discussed in section 0.

In the long run, a domestic settlement ledger may coexist and be interlinked with one or more international networks⁷. These settlement ledgers may also run on a similar ledger technology backbone. However, in view that developments in this space are still fluid, the Orchid Blueprint was not predicated on a specific platform or technology. Instead, the Orchid Blueprint focuses on determining the minimal specifications of the ledger infrastructure upon which interoperability and the safe use of digital money could be achieved. This specification is referred to as ***Orchid Compatible Ledger (OCL)*** here.

Legal and Regulatory Framework

An operator of an OCL must comply with applicable legal and regulatory requirements relevant to its conduct as an operator. The operator's compliance with these requirements will be one of the minimum specifications in the overall consideration of whether the relevant ledger infrastructure under operation will qualify as an OCL.

The operator of an OCL will need to establish comprehensive risk frameworks, policies, procedures and implement systems that enable it to identify measure, monitor and manage the range of risks that arise due to the nature of its business operations. Likewise, operators of OCL should take care to mitigate legal risks relating to the operators' business operations with respect to the OCL.

Participation

Broadening access to the settlement ledger reduces the barriers to entry and allows more entities to participate in it. Notwithstanding, there is a need to ensure that there are appropriate measures implemented by the relevant participants (refer to Table 1) to mitigate associated risks, including money laundering, terrorism financing and proliferation financing (ML/TF/PF) risks. Hence, permissionless networks whereby anyone may view, edit, and conduct any activities, including deploying smart contracts without controls or oversight are unlikely to meet the requirements of qualification as an OCL.

Accordingly, operators of settlement ledgers may decide to allow all banks and non-bank financial institutions in Singapore to participate directly in the system. However, participants will need to adhere to the platform's rulebook, which dictates the types of activities that are permissible on the platform. Every participant of the settlement ledger will be required to comply with applicable legal and regulatory requirements in respect of the activities they may conduct on the platform. For example, the operator of the settlement ledger may determine that all financial institutions may initiate transactions, but deployment of smart contracts may only be done by designated service providers.

The type of activities that each participant conducts may differ and is subject to the governance of the settlement ledger and the commercial agreement between the providers and the participants. Indirect access by retail users to settlement ledger may be allowed, if controls are in place to ensure only retail users who have undergone due diligence by a licenced financial institution may transact on the platform. In addition, entities on the settlement ledger may be required to conduct ongoing monitoring of transactions enabled through their systems to mitigate ML/TF/PF risks, to ensure that the transactions are in line with their customers' profile (and not suspicious). To limit access of specific

⁷ The publication "Enabling Open and Interoperable Networks" describes how an interlinked model may look like. (MAS, 2023)

functions to selected parties, mechanisms such as wallet address whitelisting, partitioning through subnets or sidechains, verifiable credentials, and embedded regulatory compliance checks may be employed.

Table 1: Potential participation in an Orchid Compatible Ledger (OCL)

S/N	Roles	Description
1	Consumers (e.g., customers of the service provider, retail consumers, etc.)	• Participants will go through necessary compliance check before they are given access to the settlement ledger via the wallet provider or service provider. • Participants can only utilise services (e.g., smart contracts) that are already available on the settlement ledger. • Participants cannot setup nodes to connect to the settlement ledger. • Participants cannot deploy new services (e.g., smart contracts) on to the settlement ledger.
2	Service Providers (SP) (e.g., service providers such as application, wallets, name service providers)	• SP will go through necessary compliance checks before they are given access to the settlement ledger. • SP will have to comply with additional checks and requirements to deploy smart contracts on the settlement ledger. • Depending on the service provided, SP will require the necessary licenses to be able to perform the relevant services and be subject to AML/CFT requirements, including the conduct of customer due diligence, transaction monitoring, value transfer rules, screening, etc accordingly, under an applicable legal and regulatory framework.
3	Private Digital Money Issuers (i.e., regulated stablecoin and tokenised bank liability issuers)	• Issuers will go through necessary compliance check before they are given access to the settlement ledger. • Issuers will have to comply with additional checks and requirements to deploy smart contracts on the settlement ledger. • Issuers will also need to comply with applicable regulatory requirements, similar to requirements set out in relation to the Payment Services Act and Banking Act. • Issuers may be subject to AML/CFT requirements, including the conduct of customer due diligence, transaction monitoring, value transfer rules, screening, etc.

Operations

OCLs are expected to operate 24/7, 365 days a year and facilitate immediate exchange of digital money. They need to be underpinned by best-in-class safety and security, as disruptions can adversely affect financial stability and domestic economic activities.

The management of the ledger requires the oversight from a dedicated and adequately resourced operations team. This team should be responsible for upholding the day-to-day standards and operational requirements. A strong and robust operational process would address the concern of an FMI inability to continue operations during systemic risk scenarios. Operational risks should also be outlined in the risk framework as well. Sensitive consumer and transaction data should be kept private and access to data should be kept restricted, and on a “need to know” basis.

The bulk of the usage of an OCL is likely to be made up of small value, high volume retail transactions. Therefore, an OCL needs to have sufficiently high transaction throughput. The cost of transacting also needs to be comparable or lower than existing payment infrastructure to attract and justify adoption since the large volume of transactions increases transaction cost significantly.

The provision of 24/7 availability is predicated on clearing and settlement occurring in a single step, unlike the current system in which settlement occurs with a lag. If atomic settlement is not possible, this creates downstream impacts on ensuring that associated systems can also support this capability. For example, the settlement venue itself might support 24/7 operational capability, akin to other technology systems that presently exist. However, integrations and on-ramps/off-ramps to fiat money might still be batched, to adhere to existing settlement systems’ operational hours.

Additional considerations of operating digital asset-based networks were discussed as part of the “Enabling Open and Interoperable Network paper”⁸.

4.2 Tokenisation Bridge

The tokenisation bridge describes the component that is responsible for connecting existing market infrastructure with token-based systems. The tokenisation bridge provisions digital tokens, recording them on settlement ledgers. This capability of the tokenisation bridge is provided by:

1. On-ramp and off-ramp services to account-based settlement systems.
2. Tokenisation and distribution of digital money tokens through issuance and redemption to settlement ledgers.

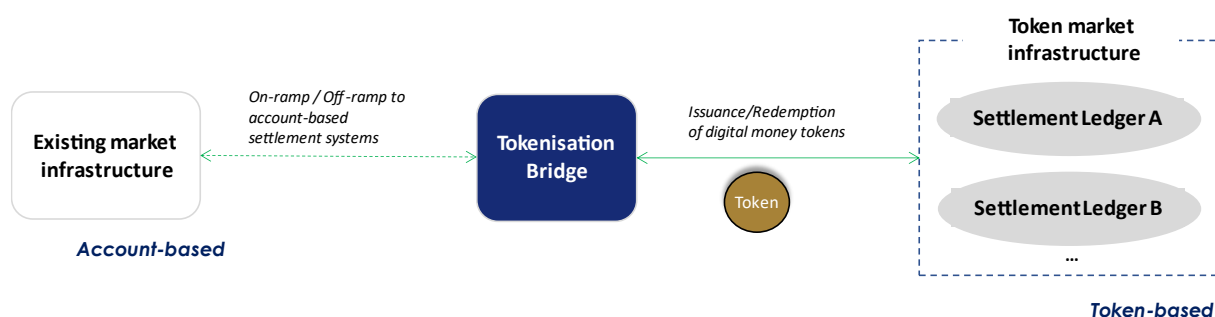


Figure 2: Tokenisation Bridge design

⁸ The publication “Enabling Open and Interoperable Networks” describes how an interlinked model may look like. (MAS, 2023)

Existing market infrastructure

Linkages with existing market infrastructures helps to ensure that users can seamlessly and safely convert between new and existing forms of money, to facilitate the adoption of digital money. This also ensures that the introduction of new forms of digital money does not lead to the fragmentation of the domestic payments system into multiple closed loop systems. “On-ramp” describes how funds in existing account-based settlement systems are exchanged for tokenised forms of money on settlement ledgers. The reverse process is termed as “off-ramp”.

The tokenisation bridge will need to ensure that these denominated values of money are accounted for accurately across the different systems. The transfers should be atomic, whereby the records are updated consistently between the source and destination systems. Notwithstanding, the connection with existing settlement systems should not impact ongoing operations supported by the existing financial infrastructure.

Token-based market infrastructure

Tokenisation bridges are designed to be compatible with different settlement ledger implementations, facilitating the issuance and redemption of digital money on these ledgers. Digital money issuers can make use of the tokenisation bridges to issue and redeem digital money tokens circulating on respective settlement ledgers.

A sample design is outlined in Figure 3, showcasing (i) a digital money smart contract that defines the rules around issuing, transferring and redeeming tokens, and (ii) a trusted wallet, which is a digital wallet that is designated by a digital money issuer for token issuance and redemption.

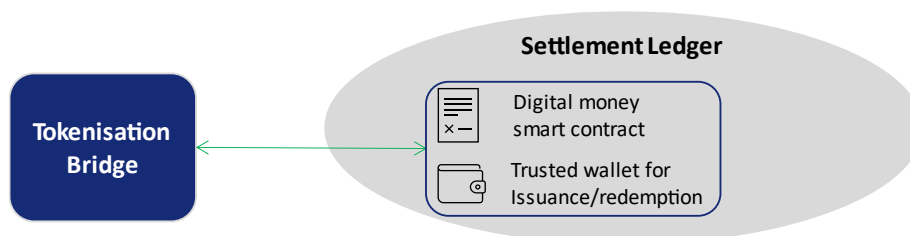


Figure 3: Sample design for connecting with settlement ledgers

For issuance, the tokenisation bridge could interact with the digital money smart contract to issue tokens to a trusted wallet that is connected to the settlement ledger. The trusted wallet can then distribute the digital money tokens to the intended recipient, through a digital money transfer. For redemption, a request can be made to redeem and transfer the digital money tokens to the trusted wallet. Upon receipt of the digital money, the tokenisation bridge could instruct the digital money smart contract to redeem the tokens from the trusted wallet.

Tokenisation bridges are designed to be platform-agnostic, and compatible with multiple types of settlement ledgers. Further, it is assumed that contractual agreements would need to be established between the tokenisation bridge provider and the destination settlement ledgers before digital tokens in the form of smart contracts can be deployed on the ledger.

Technological standards for digital money and tokenisation

Given the nascency of digital money, the methods used to represent tokenised forms of digital money are still emerging. Hence, where possible, the Orchid Blueprint takes a technology neutral, and principle-based approach to cater for future developments in digital money standards.

The two guiding principles from industry best-practices are the need for a common token standard and composability of digital money components.

Guiding principle 1: Common token standard

A common token standard helps to ensure a wide level of compatibility across different associated infrastructures, such as settlement ledgers and digital wallets. This reduces the likelihood of requiring proprietary standards and infrastructure, which create the undesired effects of technological “lock in” and fragmentation of the domestic payments network.

An example of a common token standard is the ERC-20 smart contract format, which has been widely used across different scenarios, assets, settlement ledgers and wallet implementations.

Implementing a common token standard across the different forms of digital money also reduces the technological frictions for system interaction and integration, which makes the decision to accept a particular form of digital money a governance and commercial decision, rather than one that is constrained by technological unfeasibility.

Guiding principle 2: Composability

As a system design principle, composability supports the modularity and autonomy of components. This allows for reusable components to be designed, extended, and maintained easily. In the context of digital money design, composability enables smart contracts to be developed in an efficient yet careful manner. For example, a digital money issuer might look to issue digital money onto different settlement ledgers. Instead of developing bespoke smart contracts to support each issuance, reusable smart contracts could be used to reduce duplicative work. Environment or context-specific calibrations can still be introduced through modularity, while autonomy ensures that each smart contract is neither dependent on nor impacted by calibrations made to other smart contracts.

Composability thus leads to a higher degree of freedom in developing components, reducing the barriers in developing new digital money solutions. Composability plays a large role in promoting the longevity of a technology, given the ease of extensibility, maintainability, and scalability. This is especially pertinent for digital money.

4.3 Programmability Protocol

Programmability refers to the capability to predefine a set of logic or conditions upon which activities would be executed. An example would be Purpose Bound Money (PBM)⁹, which is a standard protocol proposed by MAS that allows users to specify the conditions for the use of digital money. It is designed to work across different ledger technologies and with different forms of money.

By not programming digital money directly, the underlying digital money remains fungible once the specific purpose it was programmed for was met. In addition, by adopting PBM as a common standard, consumers using different wallets will be able to transfer digital money to one another without the need to download proprietary applications.

PBM Templates

In addition, it is envisioned that the same PBM templates for common use cases could be applied across different forms of digital money and accelerate the adoption of digital money. For example, campaign organisers could adopt a common template for creating PBM based vouchers. This allows

⁹ More information can be found in the PBM Technical Whitepaper (MAS, 2023)

merchants to accept different vouchers that are based on this common standard, without having to onboard to different schemes for each new voucher program.

The PBM templates also allow campaign organisers to programmatically add innovative conditions on top of the base templates. Similarly, for other use cases such as security deposits, financial institutions could adopt the same template while still permitting additional conditions such as partial refund, rewards, or automated conditional termination of the security deposit, to easily be embedded into the smart contract. This ensures that there is flexibility in the use of smart contracts yet ensuring assurance and composability for the exchange of digital money.

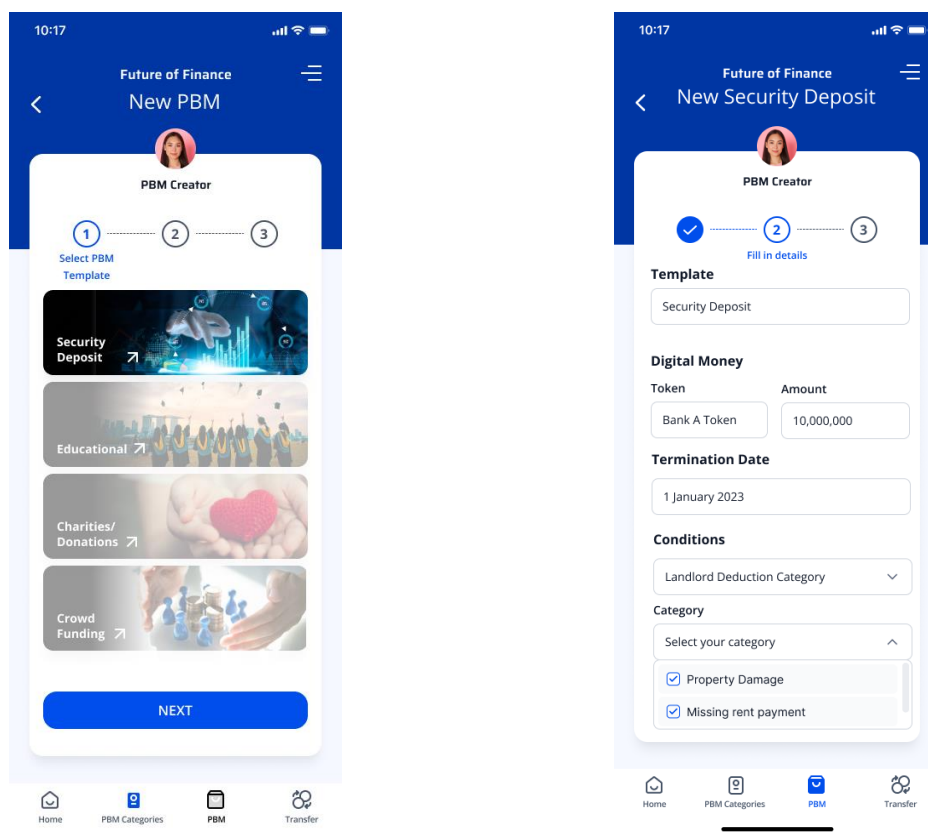


Figure 4: Illustration of a PBM Template

The PBM concept can be applied through different technologies, an example of PBM implementation is the ERC1155-Multi Token standard¹⁰. Overall, the architecture is set up to ensure that PBM protocols can adhere to robust standards, enabling versatile and secure usage in various contexts while permitting customisation and control through purpose-bound criteria.

To facilitate learning and encourage adoption, reference source code and software prototypes developed under Project Orchid have been released for public access. The collaboration among policymakers, and industry players have accelerated the collective learning and has been a catalyst for additional uses of digital money. It also further ensures a baseline of quality across different future PBM designs both domestically and internationally.

¹⁰ The Purpose Bound Money source code is available publicly on GitHub at <https://github.com/PurposeBoundMoney/PBM>

Further, the nature of PBM template shortens the time required for each PBM creation, reducing the frictions for businesses and government entities to use PBM in their digital money use cases.

Policy wrappers to facilitate transfers

Digital money transfers that are facilitated by banks undergo various controls and compliance checks, to ensure that transfers are conducted safely. These compliance checks could be implemented using PBMs, so that transfers are made only when the compliance checks are successful.

PBM which implements programmable conditions in the form of compliance, policy and regulatory checks act in essence as policy wrappers¹¹. Policy wrappers are potentially useful in digital money transfers, where the parties involved are subject to different specific policy and regulatory requirements.

For example, a commercial bank may issue TBL (refer to section Forms of Digital Money) and use it for transfers to across Know Your Customer (KYC) boundaries¹² – this is done by allowing their customers to hold and transfer the TBL to other parties that are banked by another bank that shares a trust relationship with the issuing bank. This trust relationship can be established by agreements between both banks and leverages the settlement ledger (refer to Settlement Ledger section) for a trusted reference of data.

Through the trust relationship, banks can create a “smart contract library” where the relevant smart contracts for compliance checks can be shared among banks for the creation of PBM that implements policy wrappers. This creates the context as illustrated in Figure 5.

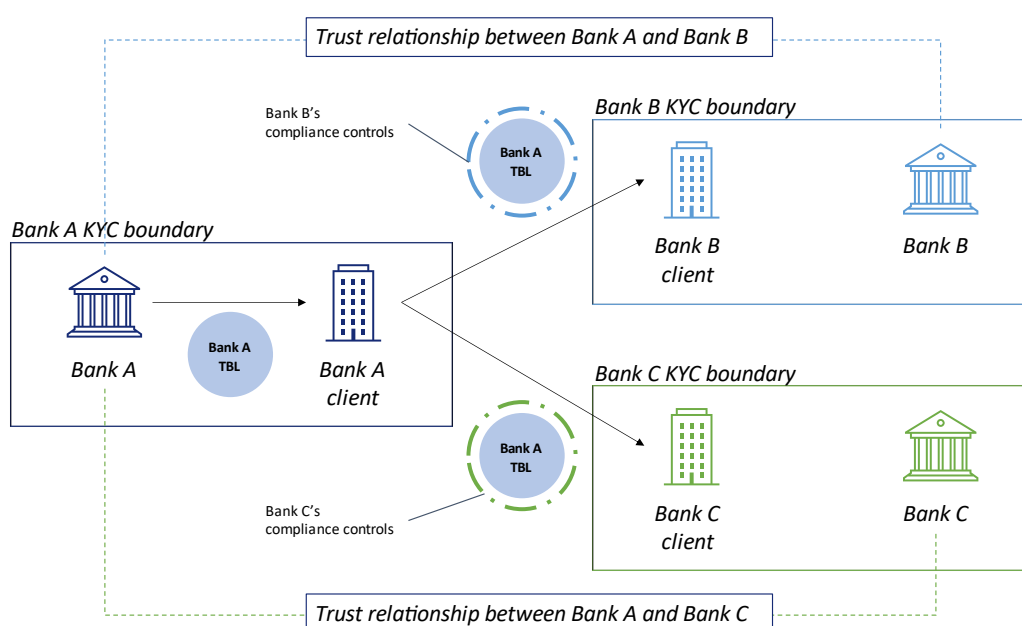


Figure 5: Embedding policy requirements through policy wrapper

As shown in Figure 5, Bank A has trust relationships with Bank B and Bank C. This creates the necessary conditions for sharing of smart contracts for compliance controls, where the validations done by one

¹¹ More details of PBM wrappers can be found in the Purpose Bound Money whitepaper.

¹² KYC boundaries describe the customer base that an issuing bank has undergone the necessary Know-Your-Customer procedures with. (BIS, 2023)

bank could be trusted and leveraged by another bank. In context, this allows Bank A's clients to be able to transfer TBL to clients of Bank B and Bank C, since the necessary policy wrappers can be provisioned, to create the PBM for the safe transfer of TBL from the perspective of Bank A. This allows the TBL to essentially cross traditional KYC boundaries.

The mechanics of creating the PBM can be seen in the illustration show in Figure 6.

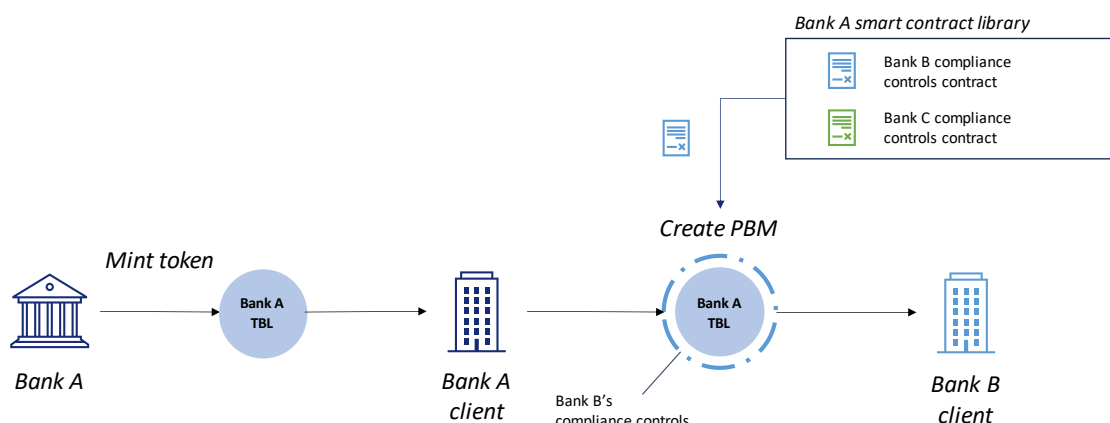


Figure 6: Creation of PBM with shared contract library

First, Bank A issues (“mints”) the Bank A TBL to Bank A client. Next, when Bank A client initiates a transfer to Bank B client, Bank B rules are applied (“create PBM”). Finally, once the rules are applied, Bank A TBL is received by Bank B client. This demonstrates a multi-institution schematic where the TBL is moving across clients of participating institutions, thus providing a reason to call additional rules.

The value of a PBM design leverages composability (as described in *Section 4.2 Tokenisation Bridge*) by allowing the modularity of programmed conditions. In the context of Figure 6, this allows Bank A to select between Bank B and Bank C conditions, as required. This is useful for the purposes of encoding policy and regulatory requirements since these requirements can be applied on a per-transaction and need-to basis. In addition, these requirements can be reused across distinct transactions, which streamlines the effort to ensure the necessary checks are performed. Given the nascency of policy wrappers, further work is required to assess its feasibility in practice. The examples provided here are meant to be illustrative only.

4.4 Name Service

Existing payment methods in the market such as PayNow¹³ rely on a set of attributes (e.g., mobile number, national identity number, business entity number, credit/debit card number, etc) to uniquely identify the destination bank account.

Meanwhile, for most token market infrastructure, the transfer of digital money is done from one wallet to another wallet. These transfers make use of the destination wallet address as the default identifier, which are a long-generated string of non-human readable characters, which is unwieldy and unmeaningful for users. This default approach of using the wallet address as the identifier is thus prone to human error and difficult for non-technological savvy users to do their verification.

¹³ More information about PayNow may be obtained from the Association of Banks in Singapore (ABS) at <https://www.abs.org.sg/consumer-banking/pay-now> (ABS, 2017)

Instead, a substitute identifier could be chosen that is meaningful to users, while reducing the difficulty in adopting this identifier. Introducing a brand-new identifier that users have no context with can incur both technological and operational effort and cost for users. On the other hand, it can be useful to leverage existing identifiers to prevent or minimise system changes required. Such an identifier may also have contextual meaning and familiarity to users, thereby improving the user experience by making use of existing associations meaningfully.

The **Orchid Name Service (ONS)** is envisioned as a service that translates a meaningful and human-readable name to a wallet address. It takes reference from the Domain Name Service (DNS), which translates a meaningful and human-readable domain name to an Internet Protocol (IP) address. ONS aims to provide a similar user experience as current payment methods and at the same time provide the ability to extend to different use cases. Figure 7 shows an example of the ONS lookup service and the information retrieved.

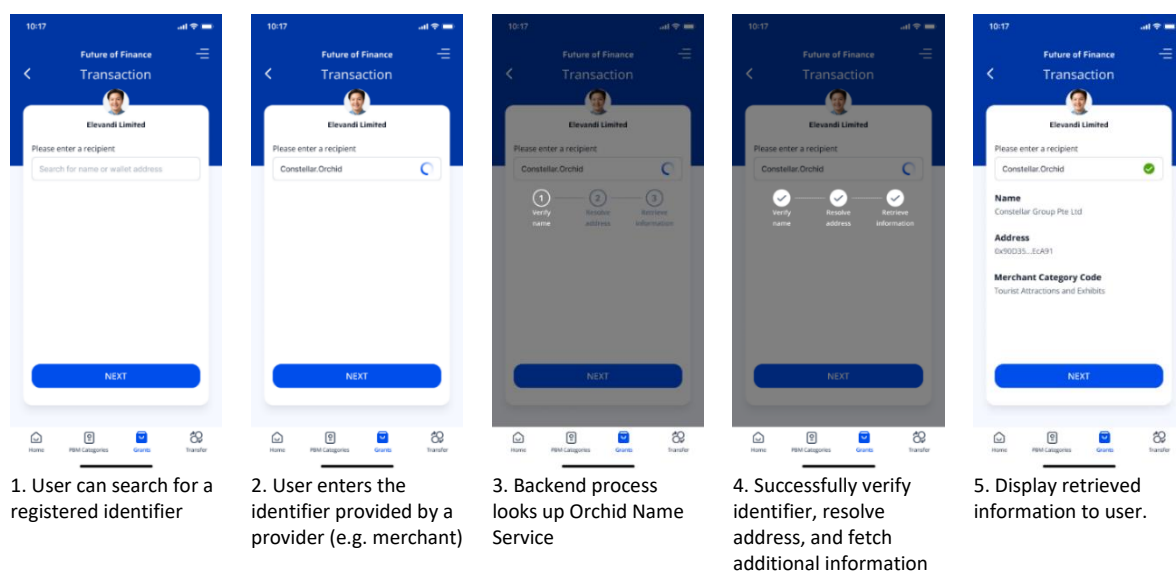


Figure 7: Orchid Name Service lookup service

Establishing and adhering to an interface standard is a key factor to foster interoperability and scalability across different implementations, applications, platforms, and ecosystems. The ONS architecture references existing domain name services such as the Ethereum Name Service (ENS)¹⁴ which maps human-readable names to machine-readable identifiers. Leveraging existing interfaces as a baseline provides the possibility to integrate and scale with existing applications that support ENS with minimal effort and cost. The ONS also references existing central addressing schemes/proxy registries (e.g., PayNow) for the supported type and format of name/identifiers. Leveraging existing name/identifiers will standardise the user experience across different solutions. The ONS interfaces can also be extended via APIs by direct participants to indirect participants with the same interface standards to allow them to also benefit from this ONS.

The ONS is made up of three components: Orchid Registry, Orchid Resolvers, and Orchid Resolver Factory.

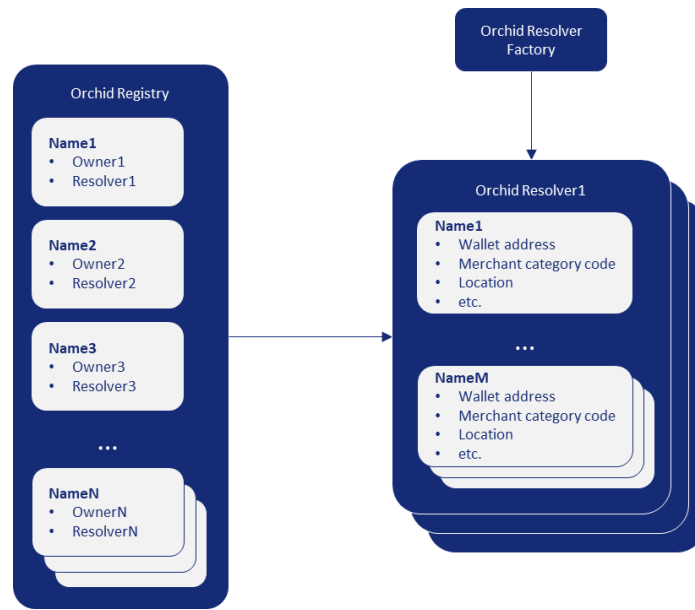


Figure 8: Orchid Name Service Architecture

1. **Orchid Registry** - Orchid Registry maintains a list of names, the owner of the name, and the resolver for resolving the name. The registry is designed to be simple and straightforward to map the name to a resolver.
2. **Orchid Resolvers** - Orchid Resolvers provide the actual translation from a name to a wallet address. It can maintain other information such as merchant category code, location, digital identity, verifiable credential, etc can also be included.
3. **Orchid Resolver Factory** - Orchid Resolver Factory is a utility that creates standard Orchid Resolver. It also acts as a registry for the list of Orchid Resolvers created.

Registration is required before users are able to make use of the ONS. During registration, users must register the desired name with the registry. Users must also provide their wallet address alongside other relevant information in the resolver. This registered name should be devised as a meaningful and readable identifier, as it will be used to link to the relevant information provided in the resolver.

On a technological level, the ONS ensures that the registered name is treated the same across different applications and systems, by following the ENS standard, which employs UTS-46 normalisation as a data normalisation step¹⁵. This data normalisation step ensures that the registered name is case-insensitive and prevents invalid characters from being input (also, serving as data cleaning). After data normalisation is completed, the normalised name undergoes a namehash¹⁶ function to generate a unique hash output. In addition, this namehash function provides some level of privacy, as the hash output that is stored does not reveal the registered name in cleartext, and the hash output cannot be reversed to obtain the registered name (i.e., one-way hashing).

¹⁵ The data normalisation describes a process of ensuring that all users get a consistent view of the ENS (ENS, 2022)

¹⁶ Namehash is a recursive function that can generate a unique hash for any valid domain name. (ENS, 2022)

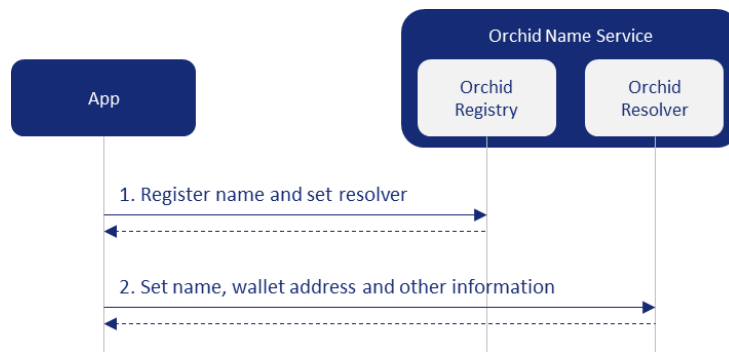


Figure 9: Register name, and set wallet address and other information

The process to resolve a name to a wallet address in ONS is a two-step process. The first step is to retrieve the resolver responsible for resolving the name, and the second step is to get the wallet address and other information. ONS can support multiple proxy/alias types such as email address, SGQRID, mobile number, UEN and ID tokens. With these proxy types, a myriad of form factors such as QR, NFC, cards and mobile apps can be used to initiate transfers. ONS can support lookup of both individuals as well as corporate profiles.

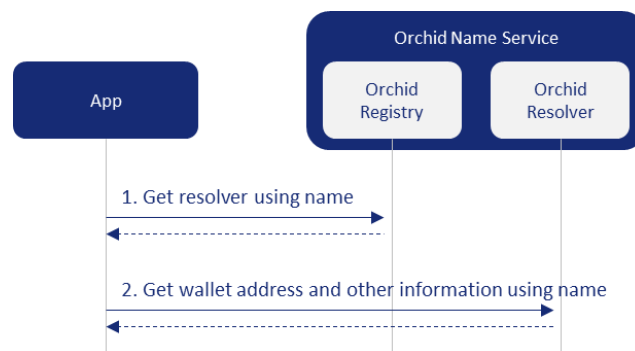


Figure 10: Resolving registered name to wallet address

Deployment Models

Entities may want to implement their own ONS and do a lookup to a peer group of ONS rather than performing a lookup to a common registry for all transactions. Under such a model, there is a need for mutual agreement between ONS providers to return the resolved address if there is a match, without any additional fee.

Such a model may be more cost-effective than paying a transaction fee to a single centralised ONS provider for each lookup. This may encourage competition among service providers and foster innovation, however, this requires coordination across different ONS providers in terms of information sharing and a common interface standard to ensure compatibility.

Meanwhile, as a merchant can have relationships with more than one acquirer, multiple resolvers may store the same name. The registry can be modified to return a list of resolvers instead of only one resolver. In this scenario, the registry should implement a function to specify the default resolver to be used when queried.

Extending ONS

ONS could be used in combination with other services to provide additional flows, logic, or conditions. For example, ONS could be extended to provide additional merchant information (e.g., MCC, location, as shown in Figure 11) which can be used as programmable conditions for PBM. ONS can link to digital identity or verifiable credential making it easier for entities to manage and share their digital identities. It can also be used for digital signatures.

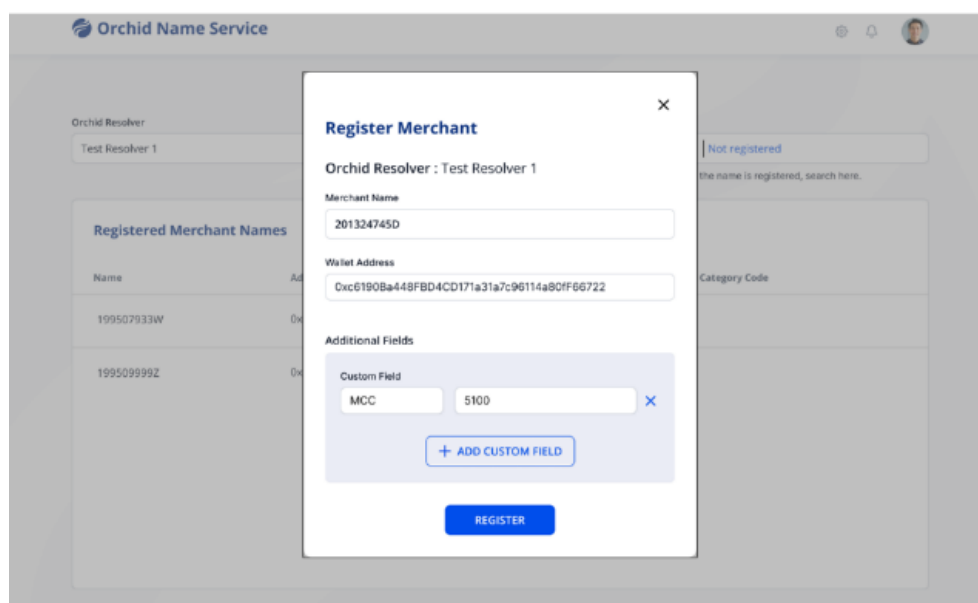
The image shows a web interface for the 'Orchid Name Service'. A modal window titled 'Register Merchant' is open. It contains the following fields: 'Orchid Resolver' (set to 'Test Resolver 1'), 'Merchant Name' (input field with '201324745D'), 'Wallet Address' (input field with 'Cxc6190Ba448FBD4CD171a31a7c96114a80f66722'), and 'Additional Fields' which includes a 'Custom Field' section with 'MCC' and '5100'. There is a '+ ADD CUSTOM FIELD' button and a 'REGISTER' button at the bottom. The background shows a 'Registered Merchant Names' table with columns for Name, Address, and Category Code.

Figure 11: Register name, wallet address and additional custom fields as identifiers

A possible way of extending the ONS would be to allow other payment systems to utilise the ONS, subject to the terms of the relevant schemes and arrangements (refer to Commercial and Operational considerations). For example, ONS supports reverse lookup where the wallet address can be used to retrieve the registered name and thus other associated information can also be retrieved. This may enable payments to take place across different payment systems, by linking together identifiers and relevant data that allow data and value to flow. As an illustration, the linkage between ONS and PayNow could be drawn, where a digital money transfer utilises ONS to resolve the wallet address, while also obtaining the relevant information of a mobile number, which can be used as the identifier in PayNow to access the FAST payment rail. This illustration shows the potentiality of cross payments.

Governance and Permissions

The registration of name and modification of information in the ONS should comply with relevant standards and requirements, including Technology and Risk Management (TRM) Guidelines. For example, Oracles¹⁷ should be selected from reputable providers with a track record of reliability and security. Additionally, ONS providers should implement access control for different functions. This is often achieved using modifiers or checks in the function code. Emergency kill-switches that allows designated administrators or owner to temporary suspend the ONS services should be implemented in case of critical issues.

¹⁷ Oracles provide a way for the decentralised Web3 ecosystem to access existing data sources, legacy systems, and advanced computations.

Safeguards to prevent abuse

To prevent abuse, the ONS can incorporate various safeguards.

At the data level, it can prevent illegal or deceitful names from being registered. The ONS could enforce a registered name specification through data normalisation and data cleaning of the registered name. This ensures that registered names do not contain any invalid characters and can also be configured to prevent unwanted words to be used as identifiers, such as expletives. In addition, names that closely resemble existing registered names could be flagged out during the data cleaning, as a potential safeguard against deceitful names.

Safeguards at the data level also extends to verification of relevant information with the resolver. When registering other relevant information with the resolver, verification checks could be done to ensure that the information is registered by the legitimate owner of the information.

The ONS can provide protections against spoofing. The usage a system font such as Consolas could help ensure that similar-looking characters are presented differently (for example, uppercase 'O' and the numeral '0' or the uppercase 'l' and lowercase 'l'). This could protect against spoofing, where an incorrect identifier that resembles the correct identifier is presented, with the intention to deceive.

The ONS can also disincentivise the use of ONS for reconnaissance. While the ONS lookup service provides relevant information when facilitating digital money transfers, this service should not be misused for reconnaissance purposes. A possible method of disincentivising such behaviour would be setting up a fee model, such that each ONS lookup requires a fee payment which is refunded/waived only when an intended function such as a digital money transfer is enacted. Another possible method involves limiting the number of lookups in a given duration.

5 User Experience considerations

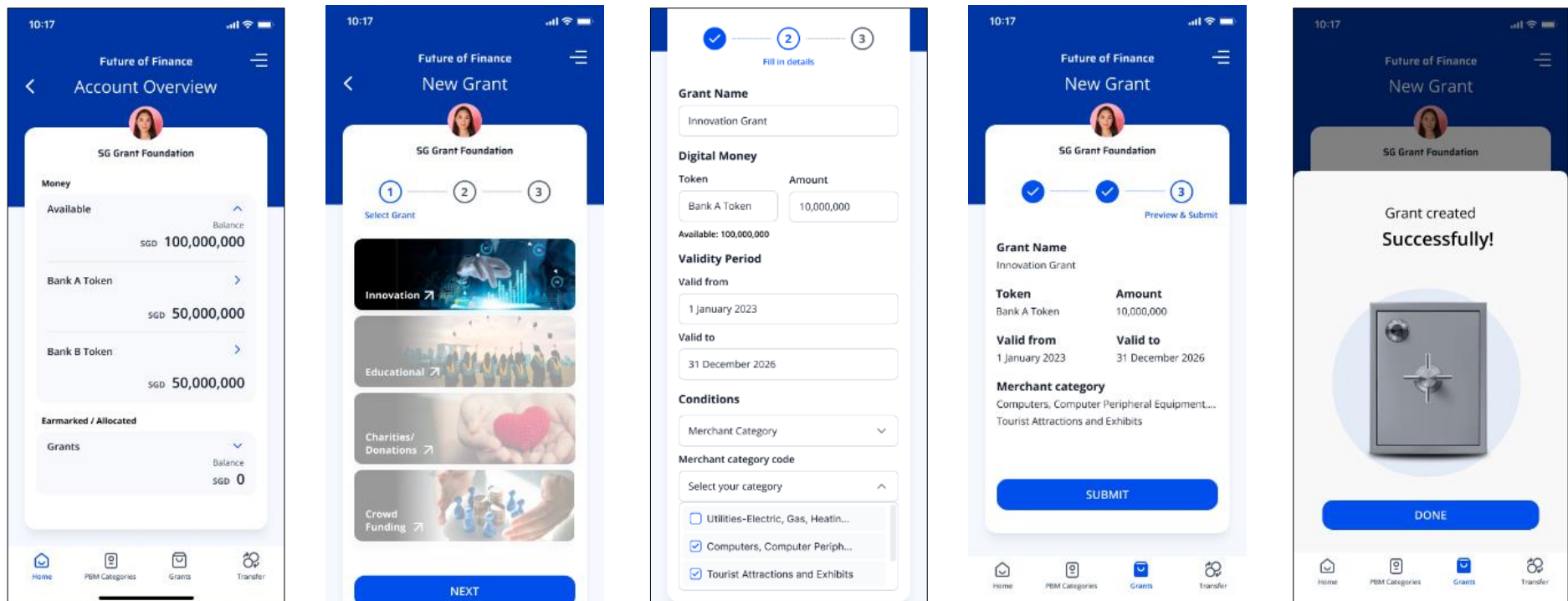
This section describes how consumers might interact with digital money through an illustrative example based on a grant disbursement use case. The example is seen from the perspectives of a grant maker, grant recipient and merchant. The section also describes the expectations on the digital wallets that would be used by users to interact with different forms of digital money.

5.1 User Journey

Grants are typically disbursed directly by government entities or through authorised entities to a person or organisation for specific purpose(s). Grants are, in numerous cases, disbursed when a user applies for a grant where money is paid out only upon after certain requirements being met. Conditions for the use of grants could be programmed to ensure grants are only disbursed to pre-identified individuals or businesses, and for a certain purpose.

In this section, an example of transactions that occur through the lifecycle of grant disbursement is illustrated— A) Grant Creation B) Grant Disbursement C) Grant Redemption D) Funds Received.

A) Grant Creation



1. SG Grant Foundation has a total of S\$100M, comprising of S\$50M worth of Bank A's Token and SGD\$50M worth of Bank B's Token. No grant has been created yet.

2. SG grant foundation selects Innovation Grants from a list of predefined templates.

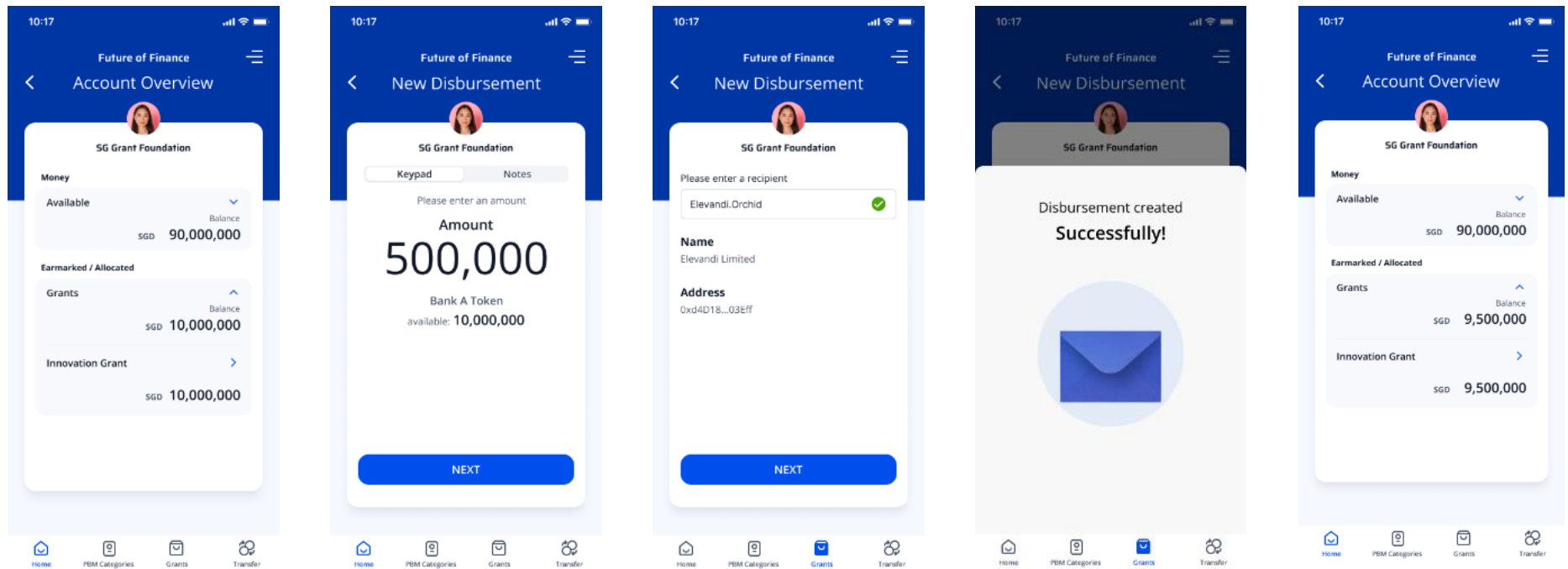
3. The grant maker defines parameters for the grant by specifying the total amount of the grant, validity period and the merchant category where the grant can be used.

4. Upon submission, the conditions are programmed with the PBM protocol.

5. Grant is now created and ready for disbursement

Figure 12: Grant Creation User Journey

B) Grant Disbursement



1. SG Grant Foundation now has S\$10M worth of Innovation grant created.

2. Out of S\$10M Innovation grant that was created, SG Grant Foundation enters S\$500K worth of grants to be disbursed to Elevandi Pte. Ltd..

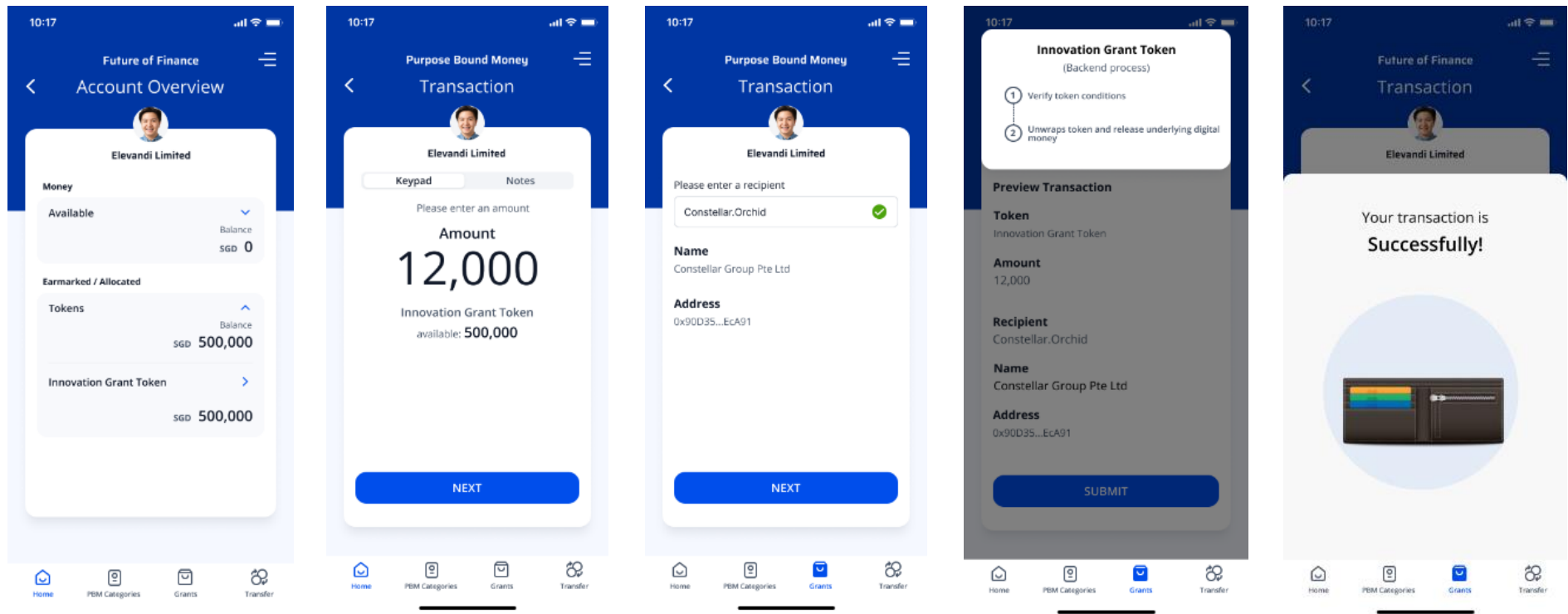
3. SG Grant Foundation specifies Elevandi.Orchid - a human readable identifier, as the Grant recipient. The Orchid Name Service will translate the identifier into a recipient's wallet address for transaction to occur.

4. Grant is now disbursed to Elevandi Limited, ready for use.

5. SG Grant Foundation will have S\$500K deducted from his account.

Figure 13: Grant Disbursement User Journey

C) Grant Redemption



1. Elevandi, as the grant recipient in this example receives S\$500K worth of Innovation Grants

2. Out of S\$500K that was received, Elevandi wants to pay S\$12K to its merchant Constellar.

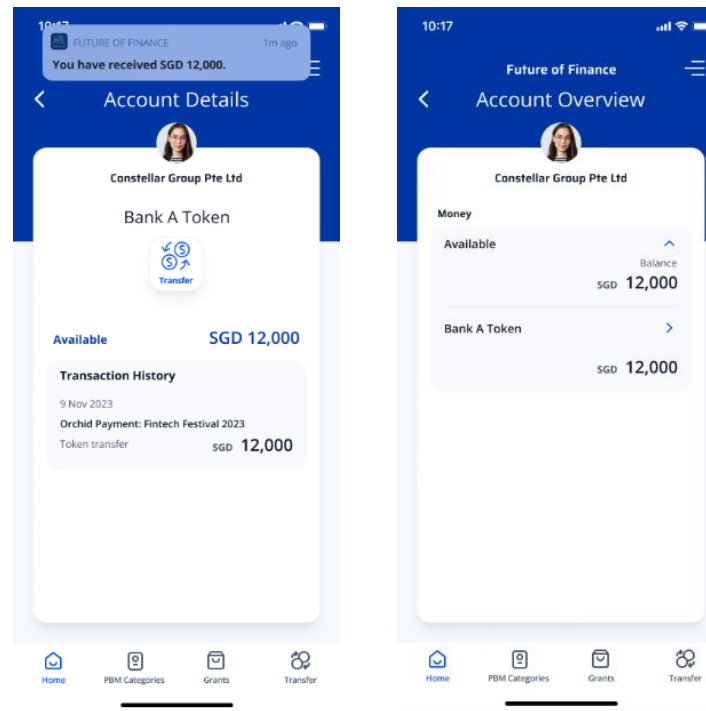
3. Elevandi specifies the recipient as Constellar.Orchid to utilise the grant.

4. Upon confirmation of the transaction, the PBM protocol verifies conditions that were initially specified by the SG grant foundation.

5. Since the conditions for use meets its pre-defined purpose, grant is utilised successfully.

Figure 14: Grant Redemption User Journey

D) Funds Received



1. Merchant receives a notification that payment has been received.

2. Merchant can view his updated balance instantly and use the money without any constraints.

Figure 15: Funds Received User Journey

5.2 Digital Wallets

As software programs that hold financial information and allow digital money transactions to be executed, digital wallets serve as the primary means that users interact with digital money.

Existing digital wallets (commonly referred to as e-wallets) store records of a consumer's account balance with a financial institution. To access the funds in the e-wallet, the consumer will need to present the login credentials provisioned by the financial institution. Any form of transfer of funds are based on instructions given by the user to the financial institution. To top up the amount of money associated with the e-wallet account, a consumer may initiate a bank transfer or by linking the e-wallet with an existing bank account or credit card facility.

Meanwhile, an emerging form of digital wallet (sometimes referred to as Web 3 wallets) provides users with the ability to hold, view and transfer digital money tokens. Web 3 wallets are made up of a pair of keys, a public and private key pair; the public key is being used as a reference addresses where tokenised forms of digital assets are stored on the ledger, while private keys corresponding to the address are used to unlock and access the user's holdings and authorise transactions. Since Web 3 wallets interact directly with the ledger, digital money tokens can be transferred to another party directly without the need to go through an intermediary or a third-party provider.

The Orchid Blueprint focuses on Web 3 wallets, in which users can send and receive digital assets (e.g. PBMs) and their respective underlying digital money.

Wallet Types

Different types of digital money may be held in different custodial wallet arrangements. A self-custodial wallet is when the private key is held and controlled by the user themselves and users can send or receive digital money without requiring additional intervention from a third party. In contrast, custodial wallets involve a third party such as a financial institution who controls and safekeeps the key on behalf of the user and acts on the user's instruction to send or receive digital money.

While some retail users may prefer to have complete control over their digital assets via self-custodial wallets, others prefer that they are held under the custody of an institution for accountability and insurance coverage purposes. The institutional setup of self-custody may require expensive hardware (e.g. Hardware Security Modules (HSM)), proper workflow systems and trained personnel to ensure compliant operations).

Since the access to the assets is controlled via a public private key pair, the form of the digital wallet is not restricted to a typical web or local application. A user can directly interact with a pre-defined logic or instructions (e.g. smart contracts) with its private key securely store in a hardware or software to execute and transfer assets.

The Orchid Blueprint describes the minimal specifications for a wallet provider to provide its services to the users. This is subject to other requirements such as the technological compatibility with key infrastructure components and the regulatory treatment of each form of digital money, which can impact the correspondent wallet arrangement.

Legal and Regulatory Compliance

A digital wallet enables users to hold on to tokenised forms of digital money to facilitate faster value transfers between corporates and individuals. The nature of bearer assets suggest that whoever holds the digital money token will have the ability to use it even if there is no direct relationship between

the issuer of the digital money and its current holder. This differs from account-based models whereby only the named owner of the account would be able to transfers funds from the account.

Before undertaking any transactions, financial institutions would need to determine the source of funds and the customer is subject to due diligence checks. Financial institutions are required to verify an individual's identity using their password and biometric data such as that described in the "MAS Guidelines on Risk Management Practices – Technology Risk".

In the case of digital assets, access is controlled via a private key. The holder of the private key can manage or transfer the digital assets using the private key, without opening an account with a financial institution beforehand. Consequently, there are concerns about the extent to which digital money as bearer tokens could facilitate nefarious activities such as money laundering or terrorism financing activities if there are no additional safeguards in place.

The wallet providers under the Orchid Blueprint framework needs to operate under the legal and regulatory jurisdiction of Singapore. Amongst others, wallet service providers are expected to comply with AML/CFT requirements as set out in MAS Notice PSN02¹⁸. This includes the need to conduct customer due diligence (including identifying and verifying the identities of the beneficial owners of the customer), perform periodic reviews of accounts, and monitor and report any suspicious transaction. On top of the compliance requirements for wallet service providers, the wallets that they provide should also be verified for use to hold and transfer digital money. Any transactions between verified and non-verified wallets are not permissible. The wallet service provider is also expected to comply with the wire or value transfer rules that set out the necessary information that would need to be provided for the transfer of funds or tokens respectively.

In accordance with the licence held by the wallet provider, the wallet should ensure that mechanisms on digital money limits (e.g. how much can be held, how much can be spent through the wallet over a given period of time) on their customers are programmed in line with requirements mandated by the regulatory authority. As an alternative, these limits could be implemented at the transaction level, where PBM is used to encode the logic for these limits when the underlying assets are transferred.

Risk

Since the digital wallets are generated via cryptographic algorithms, it is vulnerable to brute force attacks such as quantum computers. The wallet providers and/or the OCL providers have to ensure that the algorithms used to generate the keys are resistant to attacks or brute force (e.g. quantum resistant) and supported by the ledgers. OCL must provide means to allow any migration of assets from a known vulnerable algorithm generated wallet keys to a safer one if such situation occurs.

The wallet that is securing the private key must ensure measures are in placed to prevent and deter any attempts of theft, extraction and/or destruction of the keys. Any attempts to tamper the wallets must be detected and prevented by the wallet providers.

As the private key is the only means of accessing the assets, the loss of the keys would mean the loss of controls or possession of the assets. Wallet providers must educate their users about the importance of securing their keys and the risk involved, provide recovery measures to allow users to recover its keys in the event of the loss of keys (if available, e.g. through the use of Multi-party

¹⁸ The MAS Notice PSN02 defines the requirements for digital payment token service providers on anti-money laundering (AML) and countering the financing of terrorism (CFT). (MAS, 2022)

Computation (MPC¹⁹)), providing tools, resources or alternatives backup solutions to the users and ensure that the keys are indeed backed-up and that keys are recoverable.

5.3 User Protection

The proliferation of digital money and the value transfers that take place could make it a viable target for malicious attackers. Hence, measures pertaining to user protection are a necessary area of consideration. One area is strengthening the security of infrastructure, products, and services that users interact with. For example, strong and robust encryption and multi-factor authentication should be foundational. Other forms of technological detection and prevention could be helpful, such as fraud detection algorithms to identify unusual digital money transactions, which could potentially reduce the impact of material and reputational loss that a compromised user may experience.

User protection can also exist in the form of having sufficient resource and restitution. For an example, some form of customer support arrangements could be made available when users suspect they might have been compromised. Notwithstanding, there needs to be a balance between providing user protections while ensuring that they are not abused, or encourage lackadaisical behaviour among users.

¹⁹ In a general sense, MPC enables multiple parties – each holding their own private data – to evaluate a computation without ever revealing any of the private data held by each party (Source: <https://www.fireblocks.com/what-is-mpc/>) (Fireblocks, n.d.)

6 Commercial and Operational considerations

Aside from the functional capabilities of digital money, the deployment of digital money should be commercially viable and its operations manageable for its stakeholders. In this section, the commercial and operational considerations are discussed. In this section, a hypothetical scheme is used to illustrate how digital money activities under the Orchid Blueprint may be organised.

This scheme would be defined by the governing entity, and the rules that members of the scheme will be required adhere to (which set out amongst other things, the commercial arrangements applicable to participating members and their roles and responsibilities as members). It is envisioned that merchants who are onboarded by a participating acquirer will be able to receive digital money from a third party who conforms to such a scheme.

The development of the scheme is guided by the following principles:

1. **Market Efficiency** – Compatible and ease of integration to existing and upcoming banking and payment ecosystems
2. **Safety** – Assess and manages risks, ensure infrastructure are designed and operated in a safe manner that do not compromise financial stability.
3. **Ease of Use** – Minimal impact to existing merchant acceptance channels (e.g. QR) and consumer user experience
4. **Commercial Openness** – Open for existing service providers, merchant acquirers to subscribe, integrate and participate. The scheme must be scalable and technology neutral.
5. **Commercial viability for members** – The pricing must be equitable and transparent.

6.1 Roles and Responsibilities

Potential members of the scheme are listed in the table below, alongside their roles and responsibilities. Members are expected to comply with relevant and prevailing regulatory requirements, where applicable.

Table 2: Persona and Key Responsibilities

S/N	Persona	Key responsibilities
1	Private Digital Money Issuer	• Responsible for issuing digital money • Provision of redemption (off-ramp to fiat) services upon request by end users or digital money holder • Responsible for complying with applicable MAS regulations
2	Financial Service Provider	• Design campaign²⁰ and application services • Deployment of smart contract conditions based on campaign • Facilitating the campaign setup and merchant/acquirer management • Support the distribution of PBM-wrapped tokenised digital money (“PBM tokens”) to intended recipients’ wallets
3	Wallet Service Provider	• Wallet creation, custody services and user interface layer for intended users and merchants

²⁰ Campaigns refer to the sales campaign that a financial service provider might run, which are typically time-bound with campaign-specific terms of use.

		• Conduct due diligence and relevant AML/CFT measures on wallet users • Ensure that wallets can facilitate pay/receive and interoperate with existing payment infrastructures
4	Merchant Acquirer	• Merchant onboarding • Acquiring services for digital money transactions and processing • Merchant services enhancement – digital money acceptance • Facilitate merchant settlement process • Work with application service provider for campaign curation
5	Name Service Provider	• Lookup service for wallet addresses and other metadata

Providers of ancillary digital money services such as smart contract auditors, and technology providers may be regarded as associate members of the scheme.

6.2 System Context

Members of the scheme may interact with each other as illustrated in the following figure.

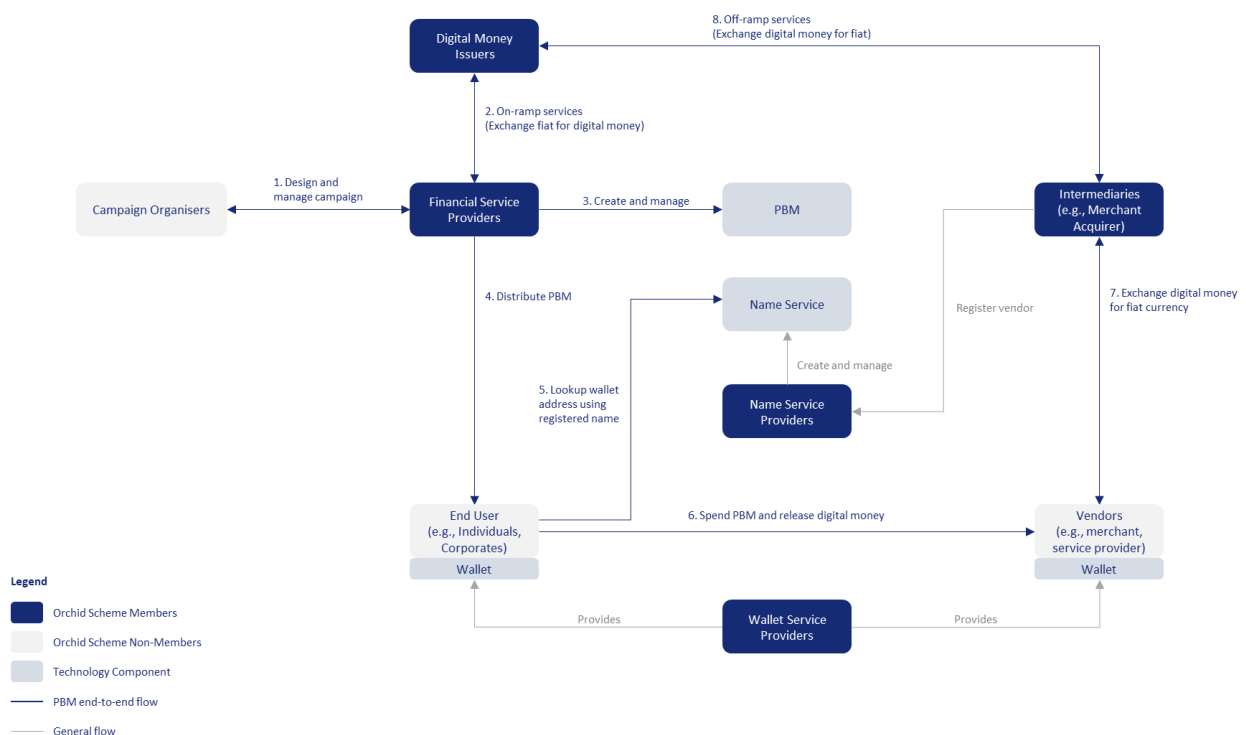


Figure 16: Interactions between different scheme members

6.3 Governing Entity

The scheme could be governed by a governing entity which will administer, enforce and prescribe rules governing members of the scheme. The governing entity may be a single entity, a group of entities, or a consortium, and may also delegate its responsibilities to a separate entity, if necessary.

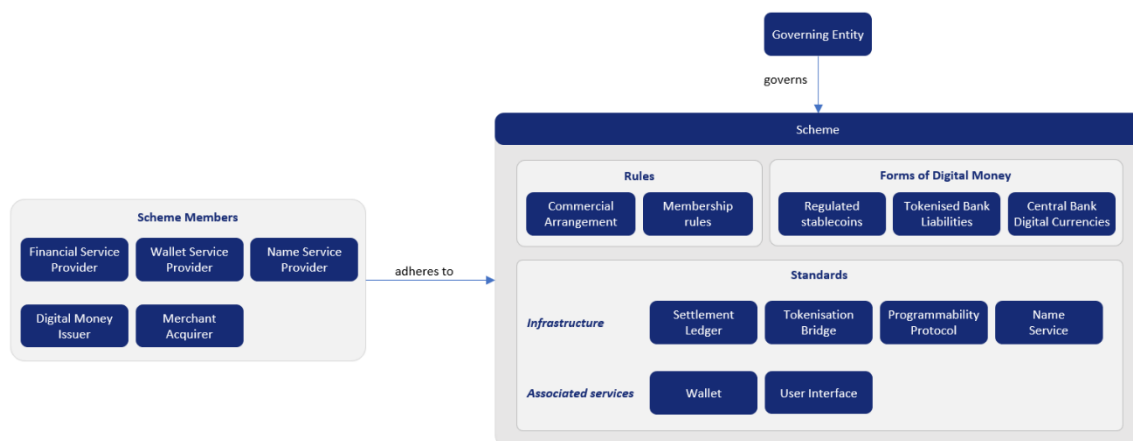


Figure 17: Scheme Overview

Entities that wish to utilise the scheme to perform selected services will need to apply to be a member of the scheme to perform those services (more details about the roles in Section 6.1 on Roles and Responsibilities). Members of the scheme will be required to comply with the rules administered by the governing entity (which may include membership and other operational rules). In addition to complying with rules administered by the governing entity, members performing regulated activities in their capacity as a scheme member would also be subject to the applicable legal and regulatory requirements in respect of that regulated activity. As a scheme member, they may be required to participate in workgroups convened to advance the development of the scheme.

6.4 Commercial Arrangement

The scheme is designed to promote the creation of open and inclusive financial infrastructures, allowing for broad-based industry participation. Accordingly, the commercial model underpinning the scheme (to be administered, enforced, and prescribed by the governing entity) should be designed in a manner that accounts for differences in separate forms of digital money issuances, be compatible with existing and emerging business ecosystems, and considers sustainability for long term adoption.

While there is no one-size-fits all approach, an efforts-based compensation allocation could represent the most sustainable path towards a long-term commercial model. This allocation recognises that not all scheme participants will perform the same responsibilities and the cost base of providing these services may differ. By clearly identifying, quantifying, and measuring these contributions, a more equitable approach could be offered. At the same time, a more transparent allocation of revenues is provided to relevant participants.

A commercial model with four baseline evaluation criteria is described below.

Table 3: Evaluation criteria

S/N	Evaluation Criteria	Principles / Rationale
1	Promote a diverse ecosystem	- Membership criteria which promote orderly entry and exit into Orchid network. - Encourage new members offering variety of digital money. - Incentivise pioneers to drive volume. - Transparent governance model.

		• Potentially scalable to cross-border use cases. • Interoperability of network infrastructure																									
2	Address pain-points of traditional fiat networks	• Reference capabilities provided by existing bank and payment systems to cater maximum use cases, while assessing gaps. • Prioritise and identify use cases which traditional systems fail to address. • Need for ongoing engagement with industry forums to review, seek inputs from stakeholders																									
3	Drive payments innovation	• Focus on network effect – grow value of the scheme with increased users. • Push-payments, Pull-payments, Request to Pay, Direct Debits <table border="1"><tr><td>Payer/Payee</td><td>Consumer</td><td>Business</td><td>Government</td></tr><tr><td rowspan="2">Consumer</td><td>C2C</td><td>C2B</td><td>C2G</td></tr><tr><td>Peer to Peer, X-border transfer</td><td>Merchant QR local/tourist</td><td>Fines, Tax, Duties</td></tr><tr><td rowspan="2">Business</td><td>B2C</td><td>B2B</td><td>B2G</td></tr><tr><td>Business payouts/payroll</td><td>Supplier Invoices</td><td>Fines, Tax, Duties</td></tr><tr><td rowspan="2">Government</td><td>G2C</td><td>G2B</td><td>G2G</td></tr><tr><td>Citizen Payouts</td><td>Govt Disbursement</td><td>Regulator exchange</td></tr></table>	Payer/Payee	Consumer	Business	Government	Consumer	C2C	C2B	C2G	Peer to Peer, X-border transfer	Merchant QR local/tourist	Fines, Tax, Duties	Business	B2C	B2B	B2G	Business payouts/payroll	Supplier Invoices	Fines, Tax, Duties	Government	G2C	G2B	G2G	Citizen Payouts	Govt Disbursement	Regulator exchange
Payer/Payee	Consumer	Business	Government																								
Consumer	C2C	C2B	C2G																								
	Peer to Peer, X-border transfer	Merchant QR local/tourist	Fines, Tax, Duties																								
Business	B2C	B2B	B2G																								
	Business payouts/payroll	Supplier Invoices	Fines, Tax, Duties																								
Government	G2C	G2B	G2G																								
	Citizen Payouts	Govt Disbursement	Regulator exchange																								
4	Equitable pricing model which is commercially viable for members	• Pricing model should cover both variable and fixed costs • variable costs (e.g. transaction fee) - pay-per-use. Considerations around underlying ledger fee treatment and fluctuations of fees. • Fixed costs (e.g. Infrastructure, licensing, central utility) - equal sharing. • Pricing principles – cost recovery-plus model. • Wallet providers and end users would need to incorporate ledger fees as part of their PBM exchange/spend journey.																									

Implementing the commercial arrangement

Commercial arrangements can be applied on infrastructure components (refer to

Key infrastructure components section) to define the membership terms and usage agreements, among other requirements. The following illustrates how the commercial arrangement can be applied to the Orchid Name Service.

Membership

By design, all scheme members will automatically be enrolled as users of the ONS. This will enable the lookup function, for wallet providers to derive the wallet address and merchant/acquirer information and use that information to initiate a transaction. An exception to the rule is if a member does not require the lookup function to execute their transactions. Taking reference from FAST and PayNow, we find that there are similar arrangements where not all FAST members are PayNow members.

The Orchid Registry of the ONS may contain participants that are not scheme members. This is in view that ONS may serve other form factors and other proxy (or alias) types, and thus it is foreseeable that ONS may be utilised for other payment schemes. The relevant data privacy and protection requirements shall apply in such an arrangement.

Operational Model

The operational model of the ONS looks at balancing operational needs, fee arrangements, ease of access and discouraging misuse of the ONS, among other considerations.

A potential operational model for the ONS could reference the DNS operational model. DNS registrars provide a domain name to IP address lookup service for free to any application, system, or person. Some registrars may offer better service levels and differentiated services for an additional fee. Similarly, ONS providers could be motivated to provide a basic lookup service to all scheme members for free, but charge a fee for additional metadata that is returned.

As the ONS lookup service is open to ONS members, a possible misuse of this service would be the mining of information in the registry for purposes other than for executing digital money transfers. To discourage abuse, providers may impose a fee structure for lookups by default, but where this fee can be waived if it leads to the execution of a digital money transfer or for the validation of PBM conditions.

7 Future Considerations

The paper has identified the key infrastructure blocks required for digital money in Singapore. Given the nascency of digital money, the infrastructure building blocks and characteristics described in this paper may evolve. In this regard, future work could include exploring the following areas.

Privacy

This paper outlines some privacy protection measures, such as the use of permissioned networks and name services. Nonetheless, individuals' transactions on the blockchain will still not be completely private. Privacy-preserving techniques often come with significant computations and storage overheads that may reduce user experience. As many of such techniques gets introduced over time, interoperability between such techniques, especially across networks, becomes more complicated as well.

Striking a balance between regulatory and compliance requirements, such as AML/CFT, are a concern for financial institutions and regulators amid worries that these techniques could facilitate illicit activities, such as money laundering and tax evasion. More ongoing work is still developing in this area and further research will be required.

Offline transactions

No IT system is infallible, especially so for digital currency ecosystems that only lives in the digital world. Offline transactions are essential for digital currencies to ensure their usability and residence in various scenarios. These capabilities enable individuals to conduct transactions even when they lack internet connectivity, during network disruptions or failure of mobile device due to low battery. Offline transactions also enhance financial inclusion by allowing less tech savvy people to participate in the digital economy. Additionally, they provide an extra layer of security, reducing the vulnerability of digital currency networks to cyberattacks and outages. In summary, offline transaction capabilities contribute to the robustness, accessibility, and security of digital currencies, making them more versatile and practical for a broader range of users and use cases.

Administration

A future consideration in the development of the Orchid Blueprint would be expanding on governance and commercial arrangements. One example would be the licencing framework. For example, whether regulators would require an analysis of each procedure to determine if the activities undertaken require a licence. The terms of such a licence, with clarity over the roles and responsibilities, are also areas for further development.

Administrative efficiency is also another area of future consideration, especially as the financial landscape for digital money matures. For example, the user onboarding journey could be streamlined and made efficient. This would be especially important in customer acquisition activities. Another consideration may come in terms of scalability, to ensure that commercial arrangements can grow, and for workloads to be efficiently managed and automated.

Cybersecurity

Cybersecurity considerations are vast and can encompass every area of digital money. In general, industry best practices should be adopted, alongside security reviews and VA/PT assessments.

Smart contracts serve an important role in the token market infrastructure. At the code level, code should be reviewed and audited according to security requirements such as OSWAP²¹. Code libraries should be selected and utilised judiciously, with proper workflows involving the remediation of vulnerabilities that arise from both direct and transitive dependencies. Secure coding practices should also be adopted to help guard against known forms of vulnerabilities and exploits such as timestamp dependency and memory leak.

²¹ The OSWAP foundation sets out security best practices for smart contracts. (Behanan & Moises, 2023)

8 Conclusion and Next steps

Advancements in technology introduce new capabilities and unlock new possibilities. For digital money to realise its potential to improve lives and bolster the economy, proper governance and rulebooks are needed to enable a safe and innovative digital ecosystem.

The industry dialogues and live pilots conducted under Project Orchid provided good opportunities to learn about the opportunities and risk of digital money and the ecosystem in which it operates in. The first phase of Project Orchid focused on uncovering the uses of digital money in Singapore, while phase two of the project examined relevant infrastructure components and commercial models.

This paper outlines the design considerations for the design and use of digital money-based services in Singapore. It considers feedback from members of the industry and introduces the Orchid Blueprint for digital money adoption. While the blueprint was developed in the context of the FinTech ecosystem in Singapore, it is envisioned that it could be extended to cater for cross-border flows as well in the future.

As the landscape continues to evolve, MAS will work with the industry group to further refine the open questions highlighted in the paper and prepare for the mainstream use of digital money in Singapore.

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10 Appendix

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